

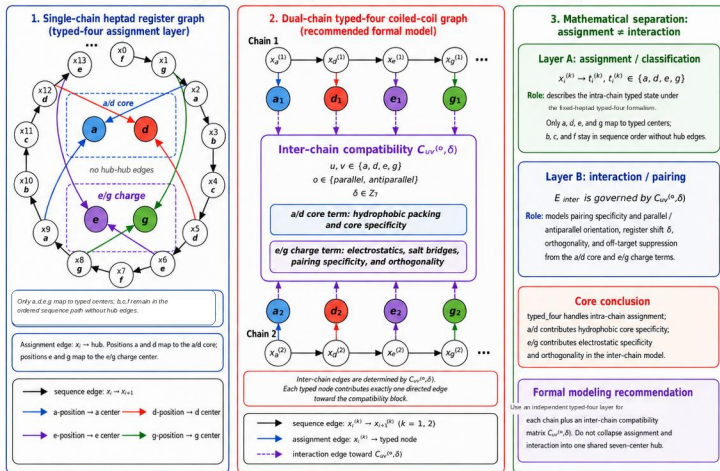
Figure 1 | Torsion-inspired typed-four dual-chain representation of coiled-coil pairs

a Open-chain cyclic phase (torsion-inspired, not a closed-knot invariant)

$$r_i^{(k)} \in \mathbb{Z}_7, \quad r_{i+1}^{(k)} = r_i^{(k)} + 1 \bmod 7, \quad \theta_i^{(k)} = 2\pi r_i^{(k)} / 7$$

Preserves cyclic order, relative phase, orientation and register displacement

b-d Typed-four assignment and dual-chain compatibility



e Computational realization: representation and learner are separated

